1. Write a console application to display "Hello, World!" using namespaces and the Main method.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Program1
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("Hello World!");
         Console.Read();
      }
   }
}
```

Output:-

```
Hello World!
```

2. Create a program that accepts command-line arguments and displays them in the console.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Program2
{
   class Program
   {
      static void Main(string[] args)
      {
         string name = args[0].ToString();
         int hour = DateTime.Now.Hour;

         if (hour >= 5 && hour < 12)
            Console.WriteLine("Good Morning " + name);
         else if (hour >= 12 && hour < 17)
            Console.WriteLine("Good Afternoon " + name);
         else if (hour >= 17 && hour < 21)
            Console.WriteLine("Good Evening " + name);
         else
            Console.WriteLine("Good Night " + name);
         Console.ReadLine();
      }
   }
}
```

Output:-

```
Good Afternoon Raju bhai
```

3. Develop a simple interactive-program to take user input (e.g., name and age) and display a formatted output.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Program3
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("Enter your name: ");
         String name = Console.ReadLine();
         Console.Write("Enter your age: ");
         int age = Convert.ToInt32(Console.ReadLine());

         Console.Write("\nName: " + name + "\nAge: " + age);
         Console.ReadLine();
      }
   }
}
```

Output:-

```
Enter your name: Raju bhai
Enter your age: 20

Name: Raju bhai
Age: 20
```

4. Write a console application to perform arithmetic operations (ADD, SUBTRACT, MULTIPLY, DIVISION, MOD).

```
Using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Project
{
   class Program
   {

      static void Main(string[] args)
      {
         Console.Write("\nCalculator \n\nEnter Number 1: ");
         int num1 = Convert.ToInt32(Console.ReadLine());
         Console.Write("Enter Number 2: ");
         int num2 = Convert.ToInt32(Console.ReadLine());

         Console.Write("Addition: " + (num1+num2) + "\n");
         Console.Write("Subtraction: " + (num1-num2) + "\n");
         Console.Write("Multiplication: " + (num1 * num2) + "\n");
         Console.Write("Division: " + (num1 / num2) + "\n");
         Console.Write("Modulus: " + (num1 % num2));
         Console.ReadKey();
      }
   }
}
```

Output:-

```
Calculator

Enter Number 1: 24
Enter Number 2: 2
Addition: 26
Subtraction: 22
Multiplication: 48
Division: 12
Modulus: 0
```

5. Create a program to declare a one-dimensional array, sort it, and display its elements. Write a program to manipulate strings (e.g., create, insert, compare, and concatenate (6-1-26)

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Program5
{
   class Program
   {
      static void Main(string[] args)
      {
         int[] array = {11,67,102,2,23,18 };
         Array.Sort(array);
         foreach (int arr in array) {
            Console.Write(arr + " ");
         }
         Console.ReadKey();
      }
   }
}
```

Output:-

```
2 11 18 23 67 102
```

6. Write a program to manipulate strings (e.g., create, insert, compare, and concatenate strings).

```
Using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Program6
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("Enter a String: ");
         string str1 = Console.ReadLine();

         Console.Write("Enter another String: ");
         string str2 = Console.ReadLine();

         int compare = string.Compare(str1,str2);
         Console.WriteLine("Compare: " + compare);

         string concatenate = string.Concat(str1,str2);
         Console.WriteLine("Concatenate: " + concatenate);

         Console.Write("\nEnter a string: ");
         string str3 = Console.ReadLine();
         Console.Write("Enter Index: ");
         int index = Convert.ToInt32(Console.ReadLine());
         Console.Write("Enter new string: ");
         string insertStr = Console.ReadLine();
         string insert = str3.Insert(index, insertStr);
         Console.Write("New String: " + insert);
         Console.ReadLine();
      }
   }
}
```

Output:-

```
Enter a String: Hello
Enter another String: World
Compare: -1
Concatenate: HelloWorld

Enter a string: Program
Enter Index: 0
Enter new string: .NET
New String: .NETProgram
```

7. Create a class Salary with Basic, TA, DA, and HRA as members. Use a constructor with default values for DA and HRA to calculate the salary of an employee.

```
Using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace Program7
{
   class Salary
   {
      int salary;
      double Ta, Da, Hra,gross_profit;
      public Salary(int salary,double Ta,double Da = 0.1,double Hra = 0.075)
      {
         this.salary = salary;
         this.Ta = Ta;
         this.Da = Da;
         this.Hra = Hra;
      }
      public void CalculateSalary()
      {
         gross_profit = salary + Ta + (Da * salary) + (Hra * salary);
         Console.WriteLine("Salary: "+salary+"\nTravel Allowance (Ta): "+ Ta+"\nDearness
Allowance (Da): "+Da);
         Console.WriteLine("House Rent Allowance (HRA): "+Hra+"\nGross: "+gross_profit);
      }
   }
   class Program
   {
      static void Main(string[] args)
```

```
        {
            Salary s1 = new Salary(15000,1000);
            s1.CalculateSalary();
            Console.ReadLine();
        }
    }
}
```

Output:-

```
Salary: 15000
Travel Allowance (Ta): 1000
Dearness Allowance (Da): 0.1
House Rent Allowance (HRA): 0.075
Gross: 18625
```

8. Demonstrate inheritance with a base class Shape and derived classes Circle, Square, and Rectangle to calculate their areas.

```
Using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Program8
{
    class Shape
    {
        public String shape;

        public Shape(String shape)
        {
            this.shape = shape;
        }

        public void display()
        {
            Console.Write("Shape: "+shape);
        }
    }

    class Circle: Shape
    {
        int radius;
        public Circle(String shape,int radius): base(shape)
        {
            base.shape = shape;
            this.radius = radius;
        }

        double area()
        {
            return 3.14 * radius * radius;
        }

        public void display()
        {
            Console.WriteLine("Shape: "+shape+"\nRadius: "+radius+"\nArea of Circle: "+area());
        }
    }
    class Square : Shape
```

```
    {
        int side;
        public Square(String shape, int side) : base(shape)
        {
            base.shape = shape;
            this.side = side;
        }

        double area()
        {
            return side * side;
        }

        public void display()
        {
            Console.WriteLine("Shape: " + shape + "\nSide: " + side + "\nArea of Square: " +
area());
        }
    }

    class Rectangle: Shape
    {
        int length,width;
        public Rectangle(String shape,int length,int width): base(shape)
        {
            base.shape = shape;
            this.length = length;
            this.width = width;
        }

        double area()
        {
            return length*width;
        }

        public void display()
        {
            Console.WriteLine("Shape: "+shape+"\nLength: "+length+"\nWidth: "+width+"\nArea
of Rectangle: "+area());
        }
    }
    class Program
    {
        static void Main(string[] args)
        {
            Shape s1 = new Shape("Triangle");
```

```
        s1.display();
        Console.WriteLine();
        Circle c1 = new Circle("Circle",2);
        c1.display();
        Console.WriteLine();
        Square sq1 = new Square("Square", 4);
        sq1.display();
        Console.WriteLine();
        Rectangle r1 = new Rectangle("Rectangle", 12,5);
        r1.display();
        Console.Read();
    }
  }
}
```

Output:-

```
Shape: Triangle
Shape: Circle
Radius: 2
Area of Circle: 12.56

Shape: Square
Side: 4
Area of Square: 16

Shape: Rectangle
Length: 12
Width: 5
Area of Rectangle: 60
```

9. Write a program demonstrating polymorphism with method overriding.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Program9
{
   class DivideByZeroExeception : Exception
   {
      public override string Message
      {
         get
         {
            return "Exception - Cannot divide by zero\n\n";
         }
      }
   }

   class CustomIndexOutOfRangeException : Exception
   {
      public override string Message
      {
         get
         {
            return "Exception - Index out of range.\n\n";
         }
      }
   }
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("Enter a Number: ");
         int num1 = Convert.ToInt32(Console.ReadLine());

         Console.Write("Enter another Number: ");
         int num2 = Convert.ToInt32(Console.ReadLine());

         try
         {
            if (num2 == 0){throw new DivideByZeroExeception();}

            double result = num1 / num2;
```

```
            Console.WriteLine("Result: " + result);
        }

        catch (DivideByZeroExeception one)
        {Console.Write(one.Message);}

        int[] arr = { 10, 20, 30 };
        Console.Write("Enter an Index (0-" + (arr.Length - 1) + "): ");
        int index = Convert.ToInt32(Console.ReadLine());

        try
        {
            if (index < 0 || index >= arr.Length)
            {throw new CustomIndexOutOfRangeException();}

            Console.Write("Value at Index " + index + ": " + arr[index]);
        }

        catch (CustomIndexOutOfRangeException one)
        {Console.Write(one.Message);}

        finally
        {
            Console.Write("\nProgram Executed Completed");
        }
        Console.ReadLine();
    }
  }
}
```

Output:-

Enter a Number: 24 Enter another Number: 2 Result: 12 Enter an Index (0-2): 3 Exception - Index out of range. Program Executed Completed	Enter a Number: 24 Enter another Number: 3 Result: 8 Enter an Index (0-2): 0 Value at Index 0: 10 Program Executed Completed	Enter a Number: 20 Enter another Number: 0 Exception - Cannot divide by zero Enter an Index (0-2): 2 Value at Index 2: 30 Program Executed Completed

10. Write a program to handle exceptions (e.g., DivideByZeroException and IndexOutOfRangeException) with try-catch and finally blocks.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Program9
{
   class DivideByZeroExeception : Exception
   {
      public override string Message
      {
         get
         {
            return "Exception - Cannot divide by zero\n\n";
         }
      }
   }

   class CustomIndexOutOfRangeException : Exception
   {
      public override string Message
      {
         get
         {
            return "Exception - Index out of range.\n\n";
         }
      }
   }
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("Enter a Number: ");
         int num1 = Convert.ToInt32(Console.ReadLine());

         Console.Write("Enter another Number: ");
         int num2 = Convert.ToInt32(Console.ReadLine());

         try
         {
            if (num2 == 0){throw new DivideByZeroExeception();}
```

```
            double result = num1 / num2;
            Console.WriteLine("Result: " + result);
        }

        catch (DivideByZeroExeception one)
        {Console.Write(one.Message);}

        int[] arr = { 10, 20, 30 };
        Console.Write("Enter an Index (0-" + (arr.Length - 1) + "): ");
        int index = Convert.ToInt32(Console.ReadLine());

        try
        {
            if (index < 0 || index >= arr.Length)
            {throw new CustomIndexOutOfRangeException();}

            Console.Write("Value at Index " + index + ": " + arr[index]);
        }

        catch (CustomIndexOutOfRangeException one)
        {Console.Write(one.Message);}

        finally
        {
            Console.Write("Program Executed Completed");
        }
        Console.ReadLine();
    }
  }
}
```

Output:-

Enter a Number: 12 Enter another Number: 2 Result: 6 Enter an Index (0-2): 2 Value at Index 2: 30 Program Executed Completed	Enter a Number: 12 Enter another Number: 0 Exception - Cannot divide by zero Enter an Index (0-2): 3 Exception - Index out of range. Program Executed Completed

11. Create a Windows Forms application to accept two numbers in Textbox, perform basic arithmetic operations, and display results in a Label. (10-2-26)

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace Program11
{
   public partial class Form1 : Form
   {
      public Form1()
      {
         InitializeComponent();
      }

      private void button1_Click(object sender, EventArgs e)
      {
         double firstNumber = Convert.ToDouble(textBox1.Text);
         double secondNumber = Convert.ToDouble(textBox2.Text);

         double add = firstNumber + secondNumber;
         label1.Text = "Addition: "+add;
      }

      private void Form1_Load(object sender, EventArgs e)
      {

      }

      private void button2_Click(object sender, EventArgs e)
      {
         double firstNumber = Convert.ToDouble(textBox1.Text);
         double secondNumber = Convert.ToDouble(textBox2.Text);

         double sub = firstNumber - secondNumber;
         label1.Text = "Subtraction: " + sub;
      }

      private void button3_Click(object sender, EventArgs e)
```

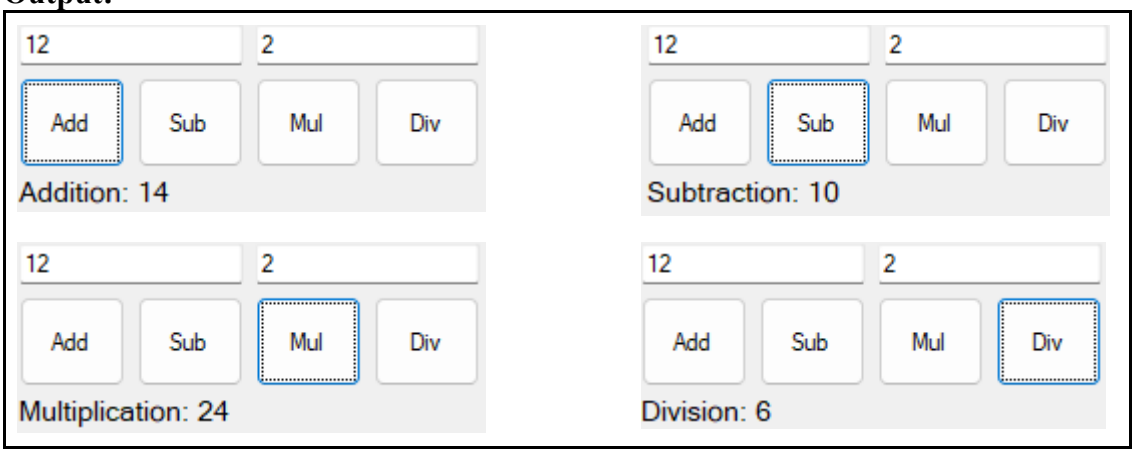
```
        {
            double firstNumber = Convert.ToDouble(textBox1.Text);
            double secondNumber = Convert.ToDouble(textBox2.Text);

            double mul = firstNumber * secondNumber;
            label1.Text = "Multiplication: " + mul;
        }

        private void button4_Click(object sender, EventArgs e)
        {
            double firstNumber = Convert.ToDouble(textBox1.Text);
            double secondNumber = Convert.ToDouble(textBox2.Text);

            if (secondNumber == 0) {
                MessageBox.Show("Error - cannot divide by zero");
            } else
            {
                double div = firstNumber / secondNumber;
                label1.Text = "Division: " + div;
            }
        }
    }
}
```

Output:-



12. Create a Windows Form application that lets a user:
1. Select an item from a ListBox (e.g., a list of destinations).
2. Choose a mode of transport from a ComboBox (e.g., Bus, Train, and Flight).
3. Pick a departure date using a DateTimePicker.
4. Submit the choices and display the selected values in a proper tabular format on new windows form. (10-2-26)

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace Program12_Lab
{
   public partial class Form1 : Form
   {
      public Form1()
      {
         InitializeComponent();
      }

      private void button1_Click(object sender, EventArgs e)
      {
         string subscription = "";

         if (Weekly.Checked)
         {
            subscription = Weekly.Text;
         }

         else if(Monthly.Checked)
         {
            subscription = Monthly.Text;
         }

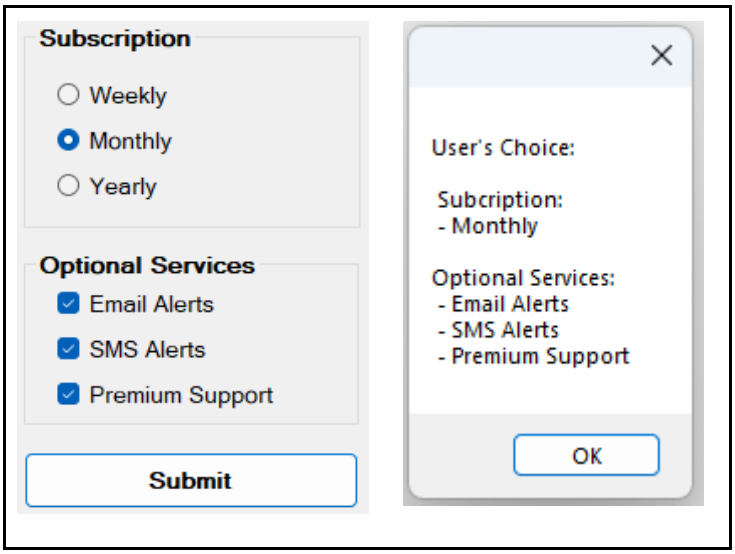
         else if (Yearly.Checked)
         {
            subscription = Yearly.Text;
         }

         List<string> services = new List<string>();
```

```
        if(checkBox1.Checked)
        {
            services.Add(checkBox1.Text);
        }
        if (checkBox2.Checked)
        {
            services.Add(checkBox2.Text);
        }
        if (checkBox3.Checked)
        {
            services.Add(checkBox3.Text);
        }

        string message = string.Join("\n - ", services);
        MessageBox.Show("User's Choice:\n\n Subcription:\n - "+subscription+"\n\nOptional
Services:\n - "+message);
    }
  }
}
```

Output:-



13. Create a Windows Form application that lets a user:
1. Select an item from a ListBox (e.g., a list of destinations).
2. Choose a mode of transport from a ComboBox (e.g., Bus, Train, and Flight).
3. Pick a departure date using a DateTimePicker.
4. Submit the choices and display the selected values in a proper tabular format on new windows form.

(form1.cs):

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace Program13
{
   public partial class Form1 : Form
   {
      public Form1()
      {
         InitializeComponent();
      }

      private void checkedListBox1_SelectedIndexChanged(object sender, EventArgs e)
      {

      }

      private void listBox1_SelectedIndexChanged(object sender, EventArgs e)
      {

      }

      private void button1_Click(object sender, EventArgs e)
      {
         if (Destinations.SelectedItem == null || comboBox1.SelectedItem == null)
         {
            MessageBox.Show("Please fill up the details.");
            return;
         }

         string destination = Destinations.SelectedItem.ToString();
```

```
            string transport = comboBox1.SelectedItem.ToString();
            string departure = dateTimePicker2.Value.ToShortDateString();

            Form2 f2 = new Form2(destination, transport, departure);
            f2.Show();
        }
    }
}
```

(form2.cs):

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace Program13
{
    public partial class Form2 : Form
    {
        public Form2(string destination,string transport,string date)
        {
            InitializeComponent();
            dataGridView1.ColumnCount = 3;
            dataGridView1.Columns[0].Name = “Destination”;
            dataGridView1.Columns[1].Name = “Transport”;
            dataGridView1.Columns[2].Name = “Date”;
            dataGridView1.Rows.Add(destination, transport, date);
        }

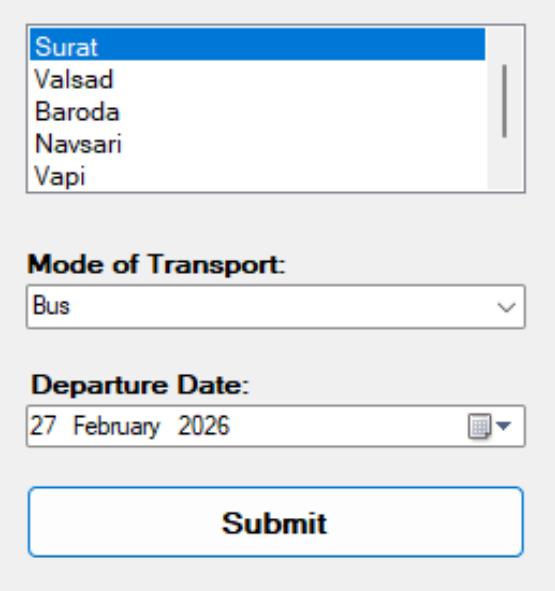
        private void Form2_Load(object sender, EventArgs e)
        {

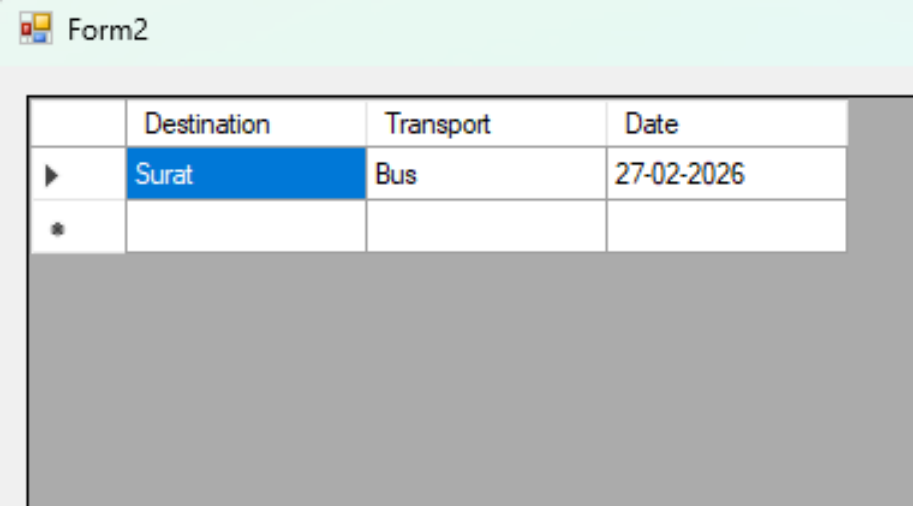
        }

        private void dataGridView1_CellContentClick(object sender, DataGridViewCellEventArgs
e)
        {

        }
    }
}
```

Output:-


Surat
Valsad
Baroda
Navsari
Vapi
Mode of Transport:
Bus
Departure Date:
27 February 2026
Submit


Form2
Destination
Transport
Date
Surat
Bus
27-02-2026

14. Develop a Windows Form application to handle mouse and keyboard events (e.g., clicking a button or entering text).
1. A Button that changes its text when clicked using the mouse.
2. A TextBox where users can type, with functionality to display the entered text in a Label upon pressing the Enter key.
3. Event handlers that detect mouse hover and key press events, showing corresponding messages.
4. A clear layout to demonstrate how user interactions are handled dynamically in a Window

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

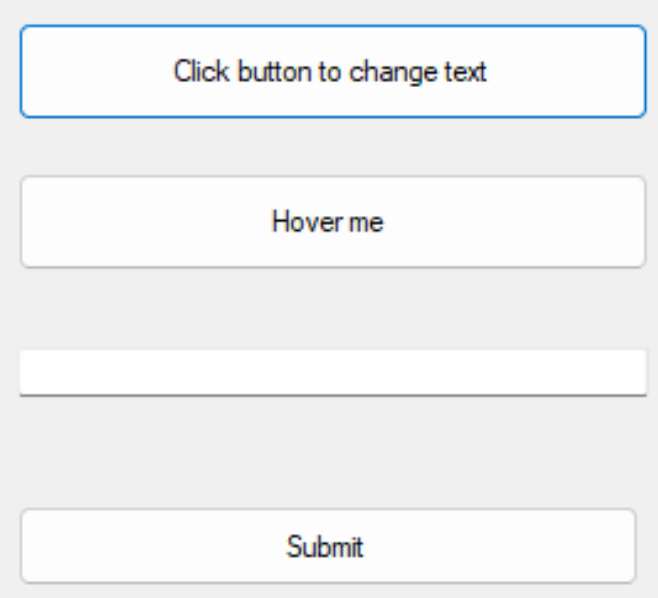
namespace Program14
{
   public partial class Form1 : Form
   {
      public Form1()
      {
         InitializeComponent();
      }

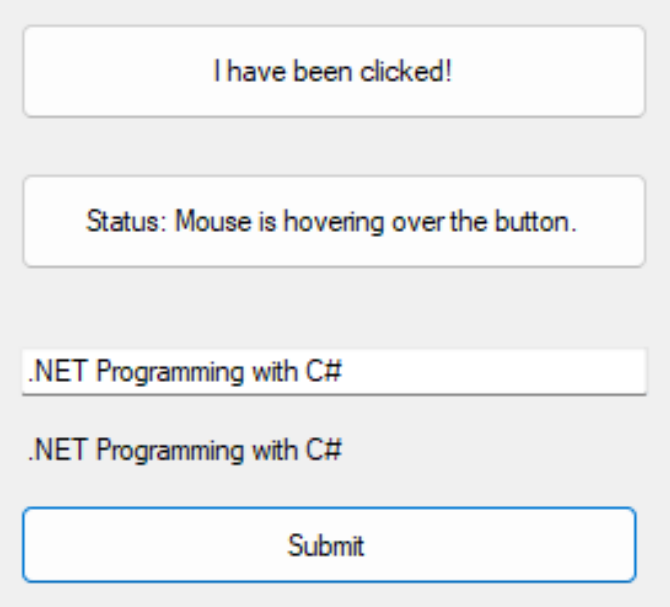
      private void button1_MouseClick(object sender, MouseEventArgs e)
      {
         button1.Text = “I have been clicked!”;
      }

      private void button2_MouseHover(object sender, EventArgs e)
      {
         button2.Text = “Status: Mouse is hovering over the button.”;
      }

      private void button3_Click(object sender, EventArgs e)
      {
         label1.Text = textBox1.Text;
      }
   }
}
```

Output:-


Click button to change text
Hover me
Submit


I have been clicked!
Status: Mouse is hovering over the button.
.NET Programming with C#
.NET Programming with C#
Submit

15. Create a simple menu-driven application with a MenuStrip and ToolStrip in Windows Forms.

```
Using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace Program15_Menustrip_and_Toolstrip
{
   public partial class Form1 : Form
   {
      public Form1()
      {
         InitializeComponent();
      }

      private void Form1_Load(object sender, EventArgs e)
      {

      }

      private void NewMenu_Click(object sender, EventArgs e)
      {
         MessageBox.Show("New command invoked.", "New", MessageBoxButtons.OK,
MessageBoxIcon.Information);
      }

      private void OpenMenu_Click(object sender, EventArgs e)
      {
         MessageBox.Show("Open command invoked.", "Open", MessageBoxButtons.OK,
MessageBoxIcon.Information);
      }

      private void ExitMenu_Click(object sender, EventArgs e)
      {
         this.Close();
      }

      private void CutMenu_Click(object sender, EventArgs e)
      {
         MessageBox.Show("Cut command invoked.", "Cut", MessageBoxButtons.OK,
MessageBoxIcon.Information);
      }

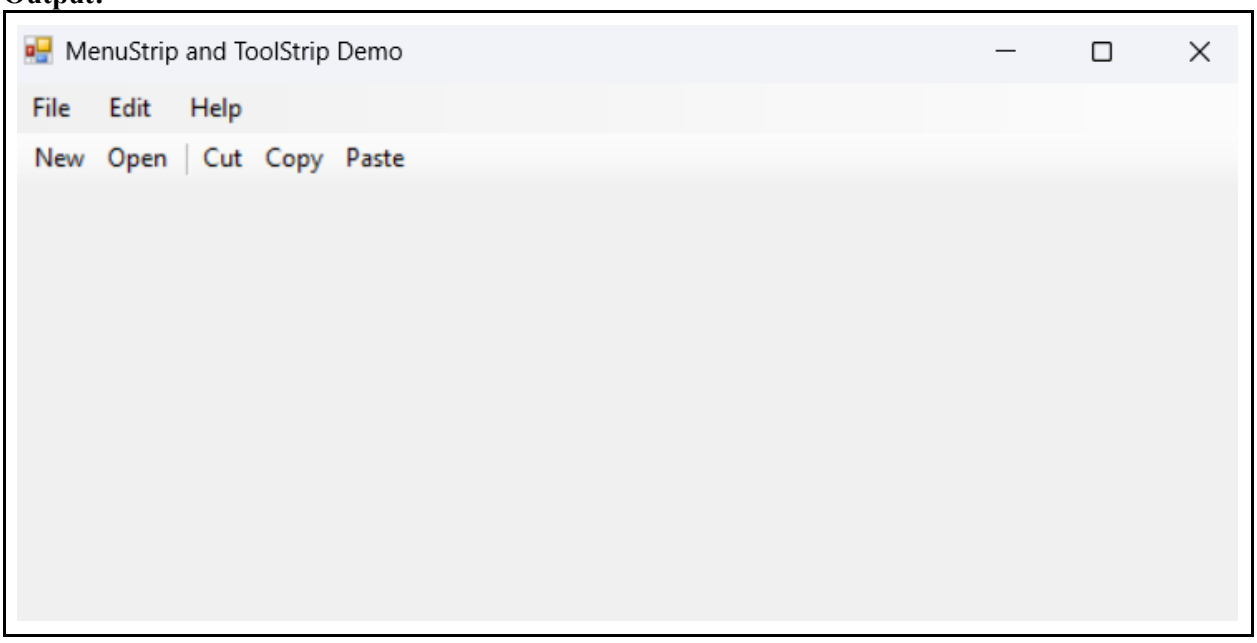
      private void CopyMenu_Click(object sender, EventArgs e)
      {
```

```
        MessageBox.Show("Copy command invoked.", "Copy", MessageBoxButtons.OK,
MessageBoxIcon.Information);
    }

    private void PasteMenu_Click(object sender, EventArgs e)
    {
        MessageBox.Show("Paste command invoked.", "Paste", MessageBoxButtons.OK,
MessageBoxIcon.Information);
    }

    private void AboutMenu_Click(object sender, EventArgs e)
    {
        MessageBox.Show("MenuStrip and ToolStrip demo\n\nSimple example application.", "About",
MessageBoxButtons.OK, MessageBoxIcon.Information);
    }
  }
}
```

Output:-



Extra Questions:

3. **Check if the input number is +,- or 0.**

```
Using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ExtraProblems
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("Enter a Number: ");
         int num = Convert.ToInt32(Console.ReadLine());
         Class1 c1 = new Class1();
         c1.CheckNumber(num);
      }
   }
}

class Class1
   {
      public void CheckNumber(int num) {
         if (num > 0)
         {
            Console.Write("Number is Positive");
         }
         else if (num < 0) {
            Console.Write("Number is Negative");
         }
         else
         {
            Console.Write("Number is Zero");
         }
         Console.ReadLine();
      }
```

Output:-

Enter a Number: 2 Number is Positive	Enter a Number: -8 Number is Negative	Enter a Number: 0 Number is Zero

2. Determine whether the given integer is odd or even.

```
Using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ExtraProblems
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("Enter a Number: ");
         int num = Convert.ToInt32(Console.ReadLine());
         Class2 c2 = new Class2();
         c2.OddEven(num);
      }
   }
}

 class Class2
   {
      public void OddEven(int num)       {

         if(num%2==0)
         {
            Console.Write("Number is Even");
         } else
         {
            Console.Write("Number is Odd");
         }
         Console.ReadLine();
      }
   }
```

Output:-

Enter a Number: 2 Number is Even	Enter a Number: 3 Number is Odd

3. Find the largest and smallest among three user inputs.

```
Using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace ExtraProblems
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("Enter Number 1: ");
         int n1 = Convert.ToInt32(Console.ReadLine());
         Console.Write("Enter Number 2: ");
         int n2 = Convert.ToInt32(Console.ReadLine());
         Console.Write("Enter Number 3: ");
         int n3 = Convert.ToInt32(Console.ReadLine());
         Class3 c3 = new Class3();
         c3.FindMaxMin(n1,n2,n3);
      }
   }
}

class Class3
   {
      public void FindMaxMin(int n1,int n2,int n3)
      {
         int largest = n1;
         int smallest = n1;
         if(n2 > largest){largest = n2;}
         if(n3 > largest){largest = n3;}
         if(n2 < smallest){smallest = n2;}
         if(n3 < smallest){smallest = n3;}
         Console.Write("\nMaximum Number: " + largest + "\nMinimum Number: " + smallest);
         Console.ReadLine();
      }
   }
```

Output:-

```
Enter Number 1: 25
Enter Number 2: 2
Enter Number 3: 100
Maximum Number: 100
Minimum Number: 2
```

4. Find the largest and smallest among 10 inputs in the array.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace ExtraProblems
{
   class Program
   {
      static void Main(string[] args)
      { Class4 c4 = new Class4();
         c4.ArrayMaxMin();
      }
   }
}
class Class4
   {
      public void ArrayMaxMin()
      {
         int[] arr = new int[10];
         for (int i = 0; i < arr.Length; i++) {
            Console.Write("arr[" + i + "]: ");
            arr[i] = Convert.ToInt32(Console.ReadLine());
         }
         Console.Write("Largest Number: " + arr.Max());
         Console.Write("Smallest Number: " + arr.Min());
         Console.ReadLine();
      }
   }
```

Output:-

```
arr[0]: 12
arr[1]: 48
arr[2]: 95
arr[3]: 100
arr[4]: 36
arr[5]: 75
arr[6]: 64
arr[7]: 34
arr[8]: 89
arr[9]: 123
Largest Number: 123
Smallest Number: 12
```

5. Check if the given number is prime or not.

```
using System;
namespace ExtraClassQuestions
{
   class Program
   {
      static void Main(string[] args)
      {
          int n;
         Console.Write("Enter Number : ");
         n = Convert.ToInt32(Console.ReadLine());

         Class5 c5 = new Class5();

         if (c5.IsPrime(n))
            Console.WriteLine($"{n} is a Prime Number.");
         else
            Console.WriteLine($"{n} is not a Prime Number.");
         Console.ReadKey();
      }
   }
}
```

```
using System;
namespace ExtraClassQuestions
{
   class Class5
   {
      public bool IsPrime(int n)
      {
         if (n <= 1)
            return false;

         for (int i = 2; i <= Math.Sqrt(n); i++)
         {
            if (n % i == 0)
                return false;
         }

         return true;
      }
   }
}
```

Output:-

Enter Number: 4
4 is not a Prime Number.

6. List all prime numbers from 1-100.

```
Using System;
namespace ExtraClassQuestions
{
   class Program
   {
      static void Main(string[] args)
      {
  Console.WriteLine("Prime numbers from 1 to 100 are : \n");

         Class7 c7 = new Class7();
         for(int I = 1;I <= 100; i++)
         {
            if (c7.IsPrime(i))
               Console.Write(I + " ");
         }
         Console.ReadKey();
          }
   }
}

using System;
namespace ExtraClassQuestions
{
   class Class7
   {
      public bool IsPrime(int n)
      {
         if (n <= 1)
            return false;
         for (int I = 2; I <= Math.Sqrt(n); i++)
         {
            if (n % I == 0)
               return false;
         }
         return true;
      }
   }
}
```

Output:-

Prime numbers from 1 to 100 are :

2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97

7. Verify if the entered number is palindrome or not.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ExtraProblems
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("Enter a number: ");
         int n1 = Convert.ToInt32(Console.ReadLine());
         Class5 c5 = new Class5();
         c5.Palindrome(n1);
      }
   }
}

class Class5
   {
      public void Palindrome(int num)
      {
         int temp, rem, res = 0;
         temp = num;
         while(num > 0)
         {
            rem = num % 10;
            res = (res * 10) + rem;
            num = num / 10;
         }

         if(temp == res){Console.WriteLine("Number is Palindrome.");}
         else{Console.WriteLine("Number is Not Palindrome.");}
         Console.ReadLine();
      }
   }
```

Output:-

Enter a number: 12321 Number is Palindrome.	Enter a number: 123 Number is Not Palindrome.

8. Verify if the entered string is palindrome or not.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ExtraProblems
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("Enter a string: ");
         string str = Console.ReadLine();

         Class6 c6 = new Class6();
         c6.StringPalindrome(str);
      }
   }
}

 class Class6
   {
      public void StringPalindrome(string str)
      {
         string reverse = new string(str.Reverse().ToArray());

         if(str == reverse)
         {
            Console.Write("String is Palindrome");
         }
         else
         {
            Console.Write("String is Not Palindrome");
         }
         Console.ReadLine();
      }
   }
```

Output:-

Enter a string: HelleH String is Palindrome	Enter a string: Hello String is Not Palindrome

9. Implement built in exceptions divided by zero index out of range or in string format. (20-1-26)

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace ExtraProblems
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("Enter dividend: ");
         int num = Convert.ToInt32(Console.ReadLine());
         Console.Write("Enter divisor: ");
         int div = Convert.ToInt32(Console.ReadLine());
         Class7 c7 = new Class7();
         c7.DivideNumber(num,div);
      }
   }
}

class DivideByZero: Exception {
      public override string Message
      {get { return "Error - number divide by zero or in string format"; }}
   }
   class Class7
   {
      public void DivideNumber(int num,int div) {
         try
         {
            if (div == 0)
            {throw new DivideByZero();}
            else{Console.Write("Result: " + (num / div));}
         }
         catch (DivideByZero one)
         {Console.Write(one.Message);\
         Console.ReadLine();
      }
   }
```

Output:-

```
         Enter dividend: 12
         Enter divisor: 0
         Error – number divide by zero or in string format
```

10. Create custom exceptions to raise an error if the age of employee is less than 18 and greater than 60.

```
Using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace ExtraProblems
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("Enter your age: ");
         int age = Convert.ToInt32(Console.ReadLine());
         Class8 c8 = new Class8();
         c8.CheckAge(age);
      }
   }
}

class AgeException : Exception
   {
      public override string Message
      {get { return "Error – age is less than 18 or greater than 60."; }}
   }
   class Class8
   {
      public void CheckAge(int age)
      {
         try
         {
            if (age < 18 || age > 60){throw new AgeException();}
            else{Console.Write("Age: " + age);}
         }
         catch (AgeException one)
         {Console.Write(one.Message);}
         Console.ReadLine();
      }
   }
```

Output:-

Enter your age: 15 Error – age is less than 18 or greater than 60.	Enter your age: 65 Error – age is less than 18 or greater than 60.

11. Create a custom exception to raise an error if the user name and password is not 'Admin' and 'admin123' respectively.

```
Using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace ExtraProblems
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("Enter username: ");
         string username = Console.ReadLine();
         Console.Write("Enter password: ");
         string password = Console.ReadLine();
         Class9 c9 = new Class9();
         c9.Login(username,password);
      }
   }
}

class LoginException : Exception
   { public override string Message
      { get { return "Error – Username or Password is incorrect."; } }
   }
   class Class9
   {
      public void Login(string username,string password)
      {
         try
         {
           if (username != "Admin" || password != "Admin123") {throw new LoginException();}
            else { Console.Write("Welcome! " + username); }
         }
         catch (LoginException one)
         { Console.Write(one.Message); }
         Console.ReadLine();
      }
   }
```

Output:-

```
Enter username: user
Enter password: adsfafa
Error - Username or Password is incorrect.
```

12. Print the result of 10 students using the structure, the structure should have roll no.student name and marks. The structure should take at least 10 records. If the student name doesn't contain alphabets then raise custom exceptions and if marks don't contain numbers then raise exceptions. This should be a menu driven program with a menu displaying all records with results, search students by roll no. and exit.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace ExtraProgram12
{
   class InvalidNameException : Exception
   {
      public InvalidNameException(string message) : base(message) { }
   }

   class InvalidMarksException : Exception
   {
      public InvalidMarksException(string message) : base(message) { }
   }

   struct Student
   {
      public int RollNo;
      public string Name;
      public int Marks;

      public string Result()
      {
         if (Marks >= 40)
         {
            return "Pass";
         }
         else {
            return "Fail";
         }
      }
   }

   class Program
   {
      static void Main(string[] args)
      {
         Student[] students = new Student[10];
```

```
for (int i = 0; i < students.Length; i++)
{
   Console.WriteLine($"\nEnter details for Student {i + 1}");

   students[i].RollNo = i + 1;

   while (true)
   {
      try
      {
         Console.Write("Enter Name: ");
         string name = Console.ReadLine();

         if (!IsAlphabetOnly(name))
            throw new InvalidNameException("Name must contain only alphabets.");

         students[i].Name = name;
         break;
      }
      catch (InvalidNameException ex)
      {
         Console.WriteLine(ex.Message);
      }
   }

   while (true)
   {
      try
      {
         Console.Write("Enter Marks: ");
         string input = Console.ReadLine();

         if (!int.TryParse(input, out int marks))
            throw new InvalidMarksException("Marks must be numeric.");

         students[i].Marks = marks;
         break;
      }
      catch (InvalidMarksException ex)
      {
         Console.WriteLine(ex.Message);
      }
   }
}
```

```
    while (true)
    {
        Console.WriteLine("\n--- MENU ---");
        Console.WriteLine("1. Display All Records");
        Console.WriteLine("2. Search Student by Roll No");
        Console.WriteLine("3. Exit");
        Console.Write("Enter choice: ");

        string choice = Console.ReadLine();

        switch (choice)
        {
            case "1":
                DisplayAll(students);
                break;

            case "2":
                SearchStudent(students);
                break;

            case "3":
                Console.WriteLine("Exiting program...");
                return;

            default:
                Console.WriteLine("Invalid choice!");
                break;
        }
    }
}
static void DisplayAll(Student[] students)
{
    Console.WriteLine("\nRollNo\tName\tMarks\tResult");
    foreach (var s in students)
    {
        Console.WriteLine($"{s.RollNo}\t{s.Name}\t{s.Marks}\t{s.Result()}");
    }
}

static void SearchStudent(Student[] students)
{
    Console.Write("Enter Roll No to search: ");
    if (int.TryParse(Console.ReadLine(), out int roll))
    {
        foreach (var s in students)
        {
```

```
                if (s.RollNo == roll)
                {
                    Console.WriteLine($"\nRoll No: {s.RollNo}");
                    Console.WriteLine($"Name: {s.Name}");
                    Console.WriteLine($"Marks: {s.Marks}");
                    Console.WriteLine($"Result: {s.Result()}");
                    return;
                }
            }
            Console.WriteLine("Student not found.");
        }
        else
        {
            Console.WriteLine("Invalid roll number.");
        }
    }
    static bool IsAlphabetOnly(string input)
    {
        foreach (char c in input)
        {
            if (!char.IsLetter(c))
                return false;
        }
        return true;
    }
  }
}
```

Output:-

Enter details for Student 1 Enter Name: A Enter Marks: 10	--- MENU --- 1. Display All Records 2. Search Student by Roll No 3. Exit Enter choice: 1 RollNo Name Marks Result 1 A 10 Fail 2 B 20 Fail 3 C 30 Fail 4 D 40 Pass 5 E 50 Pass 6 F 60 Pass 7 H 70 Pass 8 J 80 Pass 9 I 90 Pass 10 O 100 Pass	Enter choice: 2 Enter Roll No to search: 4 Roll No: 4 Name: D Marks: 40 Result: Pass

13. Create base class Person with:
1. Data member: name & age
2. Method getPersonDetails()
3. Method displayPersonDetails()

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace ExtraQs13
{
   class Person
   {
      protected string name;
      protected int age;
      public void getPersonDetails()
      {
         Console.Write("Enter Name : ");
         name = Console.ReadLine();
         Console.Write("Enter Age : ");
         age = int.Parse(Console.ReadLine());
      }

      public void displayPersonDetails()
      {
         Console.WriteLine("\nPerson Details");
         Console.WriteLine("Name : " + name);
         Console.WriteLine("Age : " + age);
      }
   }
   class Program
   {
      static void Main(string[] args)
      {
         Person p = new Person();
         p.getPersonDetails();
         p.displayPersonDetails();
         Console.ReadKey();
      }
   }
}
```

Output:-

```
Enter Name : Raju
Enter Age : 20

Person Details
Name : Raju
Age : 20
```

14. Create derived class Student that inherits from Person and contains:
1. Data member: roll no & course
2. Method getStudentDetails()
3. Method displayStudentDetails()

In Main() method:
1. Create object of Student
2. Accept and display all details

```
using System;

class Person
{
   public string name;
   public int age;

   public void getPersonDetails()
   {
      name = Console.ReadLine();
      age = Convert.ToInt32(Console.ReadLine());
   }

   public void displayPersonDetails()
   {
      Console.WriteLine(name);
      Console.WriteLine(age);
   }
}

class Student : Person
{
   public int rollno;
   public string course;

   public void getStudentDetails()
   {
      getPersonDetails();
      rollno = Convert.ToInt32(Console.ReadLine());
      course = Console.ReadLine();
   }

   public void displayStudentDetails()
   {
      displayPersonDetails();
      Console.WriteLine(rollno);
      Console.WriteLine(course);
   }
```

```
}

class Program
{
   static void Main()
   {
      Student s = new Student();
      s.getStudentDetails();
      s.displayStudentDetails();
   }
}
```

Output:-

```
Prem
19
25
IMCA
```

15. Create class Employee with constructor to set the name and display employee name.

```
using System;
namespace EmployeeApp
{
   class Employee
   {
      string name;
      public Employee(string empName)
      {
         name = empName;
      }
         public void Display()
      {
         Console.WriteLine("Employee Name: " + name);
      }
   }
   class Program
   {
      static void Main(string[] args)
      {
         Employee emp1 = new Employee("Prem");
         emp1.Display();

         Console.ReadLine();
      }
   }
}
```

Output:-

```
Employee Name: Raju
```

16. Create class Student with:
1. Name
2. Age
Use constructor to initialize them.
Display the student details.

```
using System;
namespace StudentApp
{
   class Student
   {
      string name;
      int age;

          public Student(string n, int a)
      {
         name = n;
         age = a;
      }

           public void Display()
      {
         Console.WriteLine("Student Name: " + name);
         Console.WriteLine("Student Age: " + age);
      }
   }

   class Program
   {
      static void Main(string[] args)
      {
         Student s1 = new Student("Preet", 20);
         s1.Display();

         Console.ReadLine();
      }
   }
}
```

Output:-

```
Student Name: Prem
Student Age: 20
```

17. Create class Car with:
1. Brand
2. Colour
Initialize using constructor.
Display the car information.

```
using System;

namespace CarApp
{
   class Car
   {
      string brand;
      string colour;
      public Car(string b, string c)
      {
         brand = b;
         colour = c;
      }
      public void Display()
      {
         Console.WriteLine("Car Brand: " + brand);
         Console.WriteLine("Car Colour: " + colour);
      }
   }

   class Program
   {
      static void Main(string[] args)
      {
         Car c1 = new Car("Toyota", "Black");
         c1.Display();
         Console.ReadLine();
      }
   }
}
```

Output:-

```
      Car Brand: Tesla
      Car Colour: Black
```

18. Create class Rectangle with:
1. Length
2. Width
Initialize using constructor.
Add method to calculate and display the area.

```
Using System;
namespace RectangleApp
{
   class Rectangle
   {
      double length;
      double width;
      public Rectangle(double l, double w)
      {
         length = l;
         width = w;
      }
      public void DisplayArea()
      {
         double area = length * width;
         Console.WriteLine("Length: " + length);
         Console.WriteLine("Width: " + width);
         Console.WriteLine("Area of Rectangle: " + area);
      }
   }
   class Program
   {
      static void Main(string[] args)
      {
         Rectangle r1 = new Rectangle(10, 5);
         r1.DisplayArea();
         Console.ReadLine();
      }
   }
}
```

Output:-

```
Length: 10
Width: 5
Area of Rectangle: 50
```

19. Write a C# Console Application to connect with SQL Server and perform CRUD operations.
Requirements:
1. Create database CollegeDB
2. Create table Student with fields:
- **a. StudentId (int, Primary Key)**
- **b. Name (varchar)**
- **c. Course (varchar)**
- **d. Marks (int)**

3. Menu Driven Program:
- **a. Insert Student Record**
- **b. Display All Records**
- **c. Update Student Marks**
- **d. Delete Student Record**
- **e. Search Student by ID**
- **f. Exit**

4. Use ADO.NET classes:
- **a. SqlConnection**
- **b. SqlCommand**
- **c. SqlDataReader**

5. Handle exceptions using try–catch.

```
using System;
using System.Data.SqlClient;

namespace ExtraQuestion19
{
   class Program
   {
      static SqlConnection con = new SqlConnection(
@"Data Source=(LocalDB)\MSSQLLocalDB;
AttachDbFilename=C:\Users\preet\source\repos\ExtraQues18DataBase\ExtraQues18DataBase\C
ollegeDB.mdf;
Integrated Security=True");
      static void Main(string[] args)
      {
         int choice;

         do
         {
            Console.WriteLine("\n--- Student Management System ---");
            Console.WriteLine("1. Insert Student Record");
            Console.WriteLine("2. Display All Records");
            Console.WriteLine("3. Update Student Marks");
            Console.WriteLine("4. Delete Student Record");
```

```
        Console.WriteLine("5. Search Student By ID");
        Console.WriteLine("6. Exit");
        Console.Write("Enter choice: ");

        choice = Convert.ToInt32(Console.ReadLine());

        switch (choice)
        {
            case 1:
                insertStudent();
                break;

            case 2:
                displayStudents();
                break;

            case 3:
                updateMarks();
                break;

            case 4:
                deleteStudent();
                break;

            case 5:
                searchStudent();
                break;

            case 6:
                Console.WriteLine("Exiting program...");
                break;

            default:
                Console.WriteLine("Invalid choice");
                break;
        }

    } while (choice != 6);
}

static void insertStudent()
{
    try
    {
        Console.Write("Enter Name: ");
        string name = Console.ReadLine();
```

```
            Console.Write("Enter Course: ");
            string course = Console.ReadLine();

            Console.Write("Enter Marks: ");
            int marks = Convert.ToInt32(Console.ReadLine());

            string query = "INSERT INTO Student(Name,Course,Marks)
VALUES(@name,@course,@marks)";
            SqlCommand cmd = new SqlCommand(query, con);

            cmd.Parameters.AddWithValue("@name", name);
            cmd.Parameters.AddWithValue("@course", course);
            cmd.Parameters.AddWithValue("@marks", marks);

            con.Open();
            int rows = cmd.ExecuteNonQuery();
            con.Close();

            Console.WriteLine(rows + " record inserted successfully.");
        }
        catch (Exception ex)
        {
            Console.WriteLine("Error: " + ex.Message);
            con.Close();
        }
    }

    static void displayStudents()
    {
        try
        {
            string query = "SELECT * FROM Student";
            SqlCommand cmd = new SqlCommand(query, con);

            con.Open();
            SqlDataReader reader = cmd.ExecuteReader();

            Console.WriteLine("\nID\tName\tCourse\tMarks");

            while (reader.Read())
            {
                Console.WriteLine(reader["StudentId"] + "\t" +
                                  reader["Name"] + "\t" +
                                  reader["Course"] + "\t" +
                                  reader["Marks"]);
```

```
            }

            reader.Close();
            con.Close();
        }
        catch (Exception ex)
        {
            Console.WriteLine("Error: " + ex.Message);
            con.Close();
        }
    }

    static void updateMarks()
    {
        try
        {
            Console.Write("Enter Student ID: ");
            int id = Convert.ToInt32(Console.ReadLine());

            Console.Write("Enter New Marks: ");
            int marks = Convert.ToInt32(Console.ReadLine());

            string query = "UPDATE Student SET Marks=@marks WHERE StudentId=@id";
            SqlCommand cmd = new SqlCommand(query, con);

            cmd.Parameters.AddWithValue("@marks", marks);
            cmd.Parameters.AddWithValue("@id", id);

            con.Open();
            int rows = cmd.ExecuteNonQuery();
            con.Close();

            if (rows > 0)
                Console.WriteLine("Marks updated successfully.");
            else
                Console.WriteLine("Student not found.");
        }
        catch (Exception ex)
        {
            Console.WriteLine("Error: " + ex.Message);
            con.Close();
        }
    }

    static void deleteStudent()
    {
```

```
    try
    {
        Console.Write("Enter Student ID: ");
        int id = Convert.ToInt32(Console.ReadLine());

        string query = "DELETE FROM Student WHERE StudentId=@id";
        SqlCommand cmd = new SqlCommand(query, con);

        cmd.Parameters.AddWithValue("@id", id);

        con.Open();
        int rows = cmd.ExecuteNonQuery();
        con.Close();

        if (rows > 0)
            Console.WriteLine("Student record deleted.");
        else
            Console.WriteLine("Student not found.");
    }
    catch (Exception ex)
    {
        Console.WriteLine("Error: " + ex.Message);
        con.Close();
    }
}

static void searchStudent()
{
    try
    {
        Console.Write("Enter Student ID: ");
        int id = Convert.ToInt32(Console.ReadLine());

        string query = "SELECT * FROM Student WHERE StudentId=@id";
        SqlCommand cmd = new SqlCommand(query, con);

        cmd.Parameters.AddWithValue("@id", id);

        con.Open();
        SqlDataReader reader = cmd.ExecuteReader();

        if (reader.Read())
        {
            Console.WriteLine("\nStudent Details:");
            Console.WriteLine("ID: " + reader["StudentId"]);
            Console.WriteLine("Name: " + reader["Name"]);
```

```
                Console.WriteLine("Course: " + reader["Course"]);
                Console.WriteLine("Marks: " + reader["Marks"]);
            }
            else
            {
                Console.WriteLine("Student not found.");
            }

            reader.Close();
            con.Close();
        }
        catch (Exception ex)
        {
            Console.WriteLine("Error: " + ex.Message);
            con.Close();
        }
    }
  }
}
```

Output:-

<table>
<tr>
<td>—— Student Management System —
1. Insert Student Record
2. Display ALL Records.
3. Update Student Record.
4. Delete Student Record.
5. Search Student By ID.
6. Exit</td>
<td>Enter choice: 1
Enter Name: Prem
Enter Course: DSA
Enter Marks: 99
Student record inserted successfully.</td>
</tr>
<tr>
<td rowspan="3">Enter choice: 2
ID Name Course Marks
1 Prem DSA 99
2 Rohan Java 78
3 Aisha Python 85
4 Karan DBMS 67
5 Meera AI 91
6 Rahul OS 73
7 Neha CN 88
8 Arjun ML 80
9 Kavya SQL 95</td>
<td>Enter choice: 3
Enter Student ID : 4 Enter New Marks : 76 Marks Updated Successfully</td>
</tr>
<tr>
<td>Enter choice: 4
Enter Student ID : 9
Student Record Deleted.</td>
</tr>
<tr>
<td>Enter choice: 5
Enter Student Id : 1
Student Details : Student Id : 1
Name: Prem
Course: DSA
Marks: 99</td>
</tr>
</table>

20. Create Calculator in Windows Form

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace Program20_Calculator
{
   public partial class Form1 : Form
   {
      public Form1()
      {
         InitializeComponent();
      }

      private void Form1_Load(object sender, EventArgs e)
      {

      }

      private void button1_Click(object sender, EventArgs e)
      {
         double add = Convert.ToDouble(textBox1.Text) + Convert.ToDouble(textBox2.Text);
         label4.Text = add.ToString();
      }

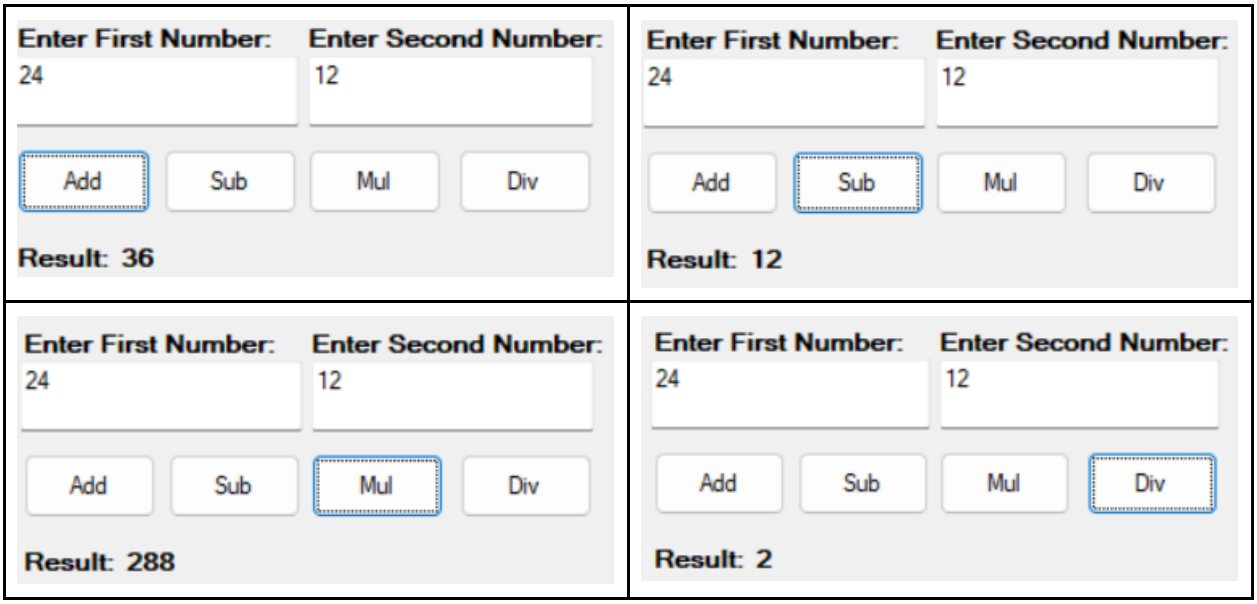
      private void button2_Click(object sender, EventArgs e)
      {
         double sub = Convert.ToDouble(textBox1.Text) - Convert.ToDouble(textBox2.Text);
         label4.Text = sub.ToString();
      }

      private void button3_Click(object sender, EventArgs e)
      {
         double mul = Convert.ToDouble(textBox1.Text) * Convert.ToDouble(textBox2.Text);
         label4.Text = mul.ToString();
      }

      private void button4_Click(object sender, EventArgs e)
      {
         double div = Convert.ToDouble(textBox1.Text) / Convert.ToDouble(textBox2.Text);
```

```
            label4.Text = div.ToString();
        }
    }
}
```

Output:-



21. Create Notepad in Windows Form with:
1. Open Dialog Box
2. Save Dialog Box
3. Print Dialog Box
4. Colour Dialog Box
5. Font Dialog Box

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Drawing.Printing;
using System.IO;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace Program21_Notepad
{
   public partial class Form1 : Form
   {
      private string _printText;
      public Form1()
      {
         InitializeComponent();
      }

      private void Form1_Load(object sender, EventArgs e)
      {

      }

      private void openToolStripMenuItem_Click(object sender, EventArgs e)
      {
         if (openFileDialog1.ShowDialog() == DialogResult.OK)
         {
            var path = openFileDialog1.FileName;
            try
            {
               if (Path.GetExtension(path).ToLower() == ".rtf")
               {
                  richTextBox1.LoadFile(path, RichTextBoxStreamType.RichText);
               }
               else
               {
                  richTextBox1.Text = File.ReadAllText(path);
               }
            }
            catch (Exception ex)
```

```
            {
                MessageBox.Show($"Error opening file: {ex.Message}", "Error", MessageBoxButtons.OK,
MessageBoxIcon.Error);
            }
        }
    }

    private void saveToolStripMenuItem_Click(object sender, EventArgs e)
    {
        if (saveFileDialog1.ShowDialog() == DialogResult.OK)
        {
            var path = saveFileDialog1.FileName;
            try
            {
                if (Path.GetExtension(path).ToLower() == ".rtf")
                {
                    richTextBox1.SaveFile(path, RichTextBoxStreamType.RichText);
                }
                else
                {
                    File.WriteAllText(path, richTextBox1.Text);
                }
            }
            catch (Exception ex)
            {
                MessageBox.Show($"Error saving file: {ex.Message}", "Error", MessageBoxButtons.OK,
MessageBoxIcon.Error);
            }
        }
    }

    private void printToolStripMenuItem_Click(object sender, EventArgs e)
    {
        printDialog1.Document = printDocument1;
        _printText = richTextBox1.Text;
        if (printDialog1.ShowDialog() == DialogResult.OK)
        {
            try
            {
                printDocument1.Print();
            }
            catch (Exception ex)
            {
                MessageBox.Show($"Print error: {ex.Message}", "Error", MessageBoxButtons.OK,
MessageBoxIcon.Error);
            }
        }
    }

    private void exitToolStripMenuItem_Click(object sender, EventArgs e)
    {
```

```
        this.Close();
    }

    private void colorToolStripMenuItem_Click(object sender, EventArgs e)
    {
        if (colorDialog1.ShowDialog() == DialogResult.OK)
        {
            richTextBox1.ForeColor = colorDialog1.Color;
        }
    }

    private void fontToolStripMenuItem_Click(object sender, EventArgs e)
    {
        if (fontDialog1.ShowDialog() == DialogResult.OK)
        {
            richTextBox1.Font = fontDialog1.Font;
        }
    }

    private void printDocument1_PrintPage(object sender, PrintPageEventArgs e)
    {
        if (string.IsNullOrEmpty(_printText))
        {
            e.HasMorePages = false;
            return;
        }

        // Measure how many characters will fit on the page
        int charsFitted = 0;
        int linesFilled = 0;
        var font = richTextBox1.Font ?? this.Font;
        var rect = e.MarginBounds;
        e.Graphics.MeasureString(_printText, font, rect.Size, StringFormat.GenericDefault, out
charsFitted, out linesFilled);

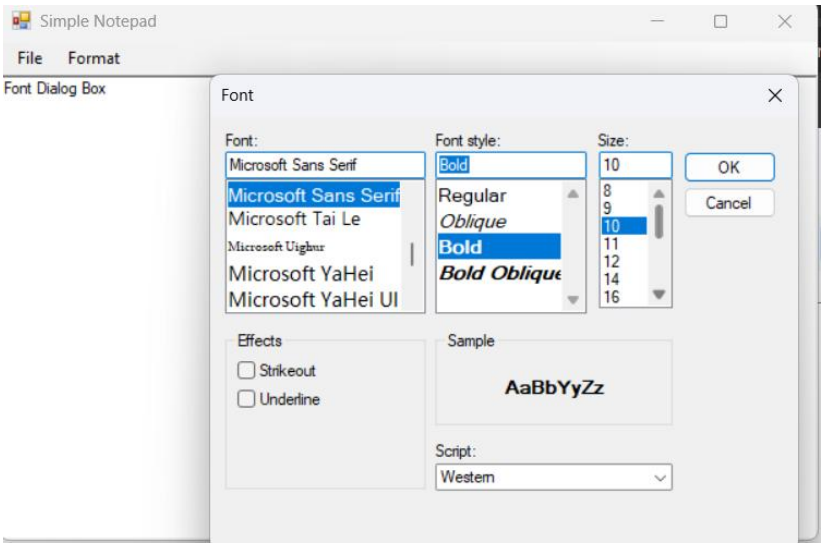
        // Draw the substring that fits
        string pageText = _printText.Substring(0, charsFitted);
        e.Graphics.DrawString(pageText, font, Brushes.Black, rect);

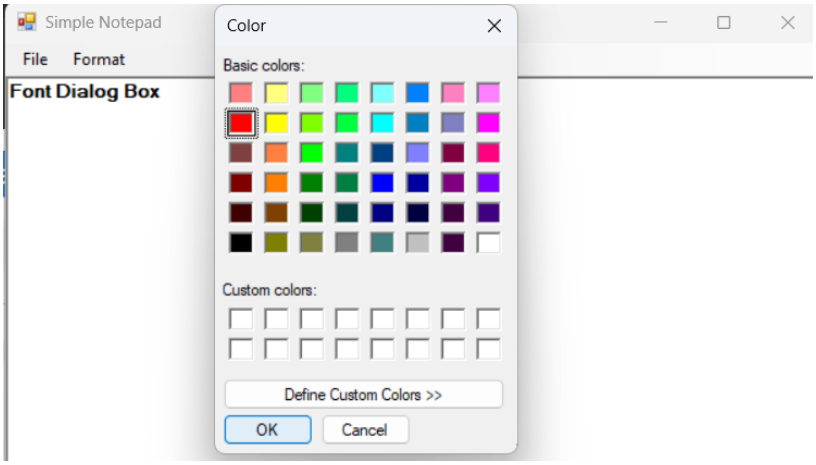
        // Remove the portion that was printed
        if (charsFitted < _printText.Length)
        {
            _printText = _printText.Substring(charsFitted);
            e.HasMorePages = true;
        }
        else
        {
            _printText = null;
            e.HasMorePages = false;
        }
    }
```

```
    }
}
```

Output:-


Simple Notepad
File
Format
Open Ctrl+O
Save Ctrl+S
Print Ctrl+P
Exit


Simple Notepad
File
Format
Font Dialog Box
Font
Font:
Microsoft Sans Serif
Microsoft Sans Serif
Microsoft Tai Le
Microsoft Uighur
Microsoft YaHei
Microsoft YaHei UI
Font style:
Bold
Regular
Oblique
Bold
Bold Oblique
Size:
10
8
9
10
11
12
14
16
OK
Cancel
Effects
Strikeout
Underline
Sample
AaBbYyZz
Script:
Western


Simple Notepad
File
Format
Font Dialog Box
Color
Basic colors:
Custom colors:
Define Custom Colors >>
OK
Cancel